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Doc. 163 – Time as Mode, Mass as Deformation, Fractality as Inside-View

Johann Pascher

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Doc. 163 – Time as Mode, Mass as Deformation, Fractality as Inside-View of Decompactified Time

Preliminary note

Doc. 162 established that T^4 is not spatially visualisable and need not be – the acceptance of mathematical constructs in physics has not depended on visualisability for over a century, but on consistency and testable consequences. That is the negative side: *what visualisation cannot deliver*.

This document makes the positive side explicit: *what an admitted inside-view metaphor can deliver*. It shows that the description of T^4 remains coherent when time is read not as a distinguished continuous variable but as a further mode on the torus – and that the fractal structure that appears in FFGFT is, in this reading, not a property of the underlying geometry but a phenomenon that arises in the transition to decompactified time. Within this transition – and only there – a particular inside-view metaphor becomes admissible, which we name explicitly and frame carefully at the end.

.1 Time as mode, not stage

On T^4 all four coordinates are equally periodic. None of them is distinguished as “time”; the division into “three spatial dimensions plus one temporal dimension” is not laid down in the carrier itself,

but arises only from the choice of a descriptive perspective. In the FFGFT reading, what we phenomenally experience as *flowing time* is a *mode* in the sense of an oscillation or motion pattern on the torus – not a stage on which physics takes place.

This position is consistent with two points already established in the corpus:

- In FFGFT, $\hbar$ is an SI conversion factor between energy and time measures, not an ontological primitive (Doc. 001a, Doc. 230 § Operational equivalence). The status of time itself is therefore not primitive either, but derived.
- The foundational relation $\tilde{T} \cdot m = 1$ is, on T^4 , a strict reciprocity, not an open time coordinate. Reciprocity is a property of a closed structure, not of a flow.

Consequence. What we perceive as “continuously flowing time” is the phenomenal resolution of a discrete torus mode into a quasi-continuous variable. This resolution we call *decompactification of time*. It is not wrong – it delivers exactly the familiar predictions of standard physics – but it is a *descriptive choice*, not a property of the carrier.

.2 Mass as deformation of the carrier

In general relativity, mass is described as curvature of the space-time manifold. On T^4 , this idea carries over structurally: what GR reads as local curvature of an open manifold becomes, on the closed carrier, a local *deformation* of the torus geometry. The global topology is preserved – the four identifications continue to close the carrier – but the local metric is altered by mass.

Three remarks:

- In the limit of small deformation (local inertial frames), the deformed torus is indistinguishable from the Euclidean space-time of GR – GR is, in this reading, the linear approximation in which the global closure is invisible.
- GR singularities (black hole, Big Bang) appear structurally differently on T^4 because the carrier is closed through identifications (cf. Doc. 230 § Operational equivalence, the point on singularities).

- The deformation is not visualisable – for exactly the reasons developed in Doc. 162. What is described here is the *structural relationship* between mass and carrier, not a picture.

.3 Fractality as an inside-view phenomenon

FFGFT contains a fractal correction structure (factor K_{frak} , appearing in π -gates and in the lepton-mass cascade). Until now this fractality has been treated in the corpus as a geometric property of T^4 . The discussion of time-as-mode shows that a more precise reading is possible:

As long as time is a discrete mode on the torus, there is no genuine self-iteration. A fractal recursion requires the application of an operation to its own result – and that requires a direction in which “before” and “after” can be distinguished. On a closed torus mode this direction does not exist. It arises only when the mode is decompactified into a directed, quasi-continuous time.

In this reading, the fractal structure is not an ontological feature of the underlying geometry but a *phenomenon of the decompactified time description*. It appears as soon as one enters the mode “directed time” – that is, as soon as one adopts an inside view in which “before” and “after” can be distinguished.

Structural parallel to the Born rule. This position is not isolated. It is structurally parallel to the Born-rule reading from Doc. 230 (§ Operational equivalence): there too, the apparent randomness of quantum measurement arises only in the transition from the deterministic T^4 dynamics to decompactified time. Both phenomena – apparent randomness and fractal inside-geometry – are consequences of the *same decompactification*, not independent findings.

This structural parallel is a non-trivial result: two phenomena that stand unconnected in standard physics (quantum randomness and fractal corrections) appear in FFGFT as two manifestations of a single cause.

.4 Inside-view metaphor: brain folds and light

Preliminary note on admissibility. The following metaphor is introduced here *explicitly*, because an openly named metaphor remains controllable, while a hidden allusion invites suspicion of esotericism. It is an *inside-view metaphor*, not an ontological claim about T^4 . It does not say " T^4 looks like a brain"; it says something more specific and weaker.

The metaphor. Once time is decompactified, an observer inside the geometry experiences not the simple structure of the carrier but a fractally folded experiential geometry – with folds, fold-curvatures, and self-similarities on several scales. The folded structure of a brain – many-times wound, self-similar, geometrically rich on a small volume – is a familiar image for exactly this kind of inside structure. It is not *what* T^4 is, but it is an appropriate visualisation of what the *inside view of decompactified time* sees.

The light example sharpens the point. Viewed from outside – in the undistorted description – light moves in a straight line. Within the deformed geometry, however, the path actually traversed is different: it follows the folds and windings, and for the observer situated inside, *this* winding path is the real one, not the idealised straight line of the outside description. The standard reading – straight light paths in Euclidean space – is, in this view, the *outside description*, which holds in the limit of undistorted geometry; the inside view is the description in which we actually live and think.

The clean framing – what the metaphor delivers and what not.

- *What it delivers:* It illustrates that a fractally complex inside-experiential world is compatible with a simple outside carrier structure. It makes plausible why our phenomenal world *appears* complex while its carrier is simple.
- *What it is not:* It is not a claim that the universe is "like a brain", or that consciousness is "fractal cosmic structure", or anything similar. Such statements would overstep the frame from metaphor to ontology and would not be defensible within FFGFT.
- *Where it applies:* Only in the transition "discrete mode → decompactified time". Outside this transition – in the pure T^4 description – there is no fractal inside-geometry to describe. The metaphor applies to the inside view, not to T^4 itself.

Within this delimitation the metaphor is methodologically admissible and didactically valuable. It closes a gap that Doc. 162 left open: Doc. 162 showed what *no* form of visualisation can deliver (spatial intuition); this document shows what a *deliberately delimited* inside-view metaphor can deliver (rendering plausible the relationship between a simple outside structure and a complex inside world).

.5 Consequences

From the four preceding sections, three statements follow that individually already stand in the corpus, but are brought together here for the first time as a common structural line:

1. **Time is not distinguished.** It is a mode among four equally periodic coordinates on T^4 . Phenomenal time-experience is the inside view of decompactified time.
2. **Mass is deformation of the carrier.** What GR describes as spacetime curvature is, on T^4 , a local deformation of the closed geometry – with standard GR as the linear, locally inertial approximation.
3. **Fractality is an inside-view phenomenon.** The fractal structure that FFGFT sees appearing in K_{frak} , in the lepton-mass cascade, and in the π -gates is a phenomenon of decompactified time, not a property of the carrier itself. Parallel to the Born-rule reading from Doc. 230, two otherwise unconnected phenomena – quantum randomness and fractal corrections – thus appear as two manifestations of the same cause.

Methodological position. This document tightens no empirical prediction and changes no calculation. It is a *reading clarification*: the structural relationship between the T^4 carrier, decompactified time, and phenomenal experiential world is made explicit. The practical use is that the metaphor of brain folds – and similar inside-view images – now have a cleanly framed standing in the corpus and need no longer operate as unexplained allusions.

.6 Summary

On T^4 , time is not a stage but a mode. Mass is not an object on a spacetime but a deformation of the carrier. Fractality is not a property of the carrier but a phenomenon arising in the transition to decompactified time – structurally parallel to the Born rule and the apparent randomness of quantum measurement. Within this transition the brain-folds metaphor is an admissible inside-view visualisation – not as a claim about T^4 , but as an image for how a simple outside structure becomes a fractally wound inside world.

The outside description is simple. The inside world in which we live is wound. The two descriptions are not in contradiction – they are two sides of the same transition.